



Department of Electrical Electronics Engineering Academic Year 2024 – 2025 (Odd Semester)

Degree, Semester & Branch: B.E/V/EEE

Course Code & Title: EE3032/Energy Storage Systems

Name of the Faculty member (s): Mrs G.Sivapriya

Innovative Practice Description

- **Unit / Topic:** Unit I / Types of Energy Storage Systems
- **Course Outcome:** CO 1
- **Topic Learning Outcome:** TLO2
- **Activity Chosen:** One Minute Paper
- **Justification:**

The basics of Energy storage Systems are discussed. After teaching this topic, I thought of this activity for making to have clarity about the various types of energy storage systems. I can judge the understanding level of the students.

Time Allotted for the Activity: 5 Minutes

- **Details of the Implementation:**

After teaching the concept, I gave students one or two minutes to think about the topic without writing anything.

Total Strength :63

Reporter: Myself

At the end of the Class (Last 5 minutes)

I asked the students to think various types of energy storage systems concept for 3 minutes.

Then I told them to write as much as they can within a short period of time (1 minute)

Finally, I collected the papers from each column

- **CO – PO / PSO mapping:**

CO	PO1	PO5	PO6	PO7	PO8	PO9	PO10	PO12	PSO1	PSO2	PSO3
CO1	2	2	3	3	1	3	3	2	2	1	1

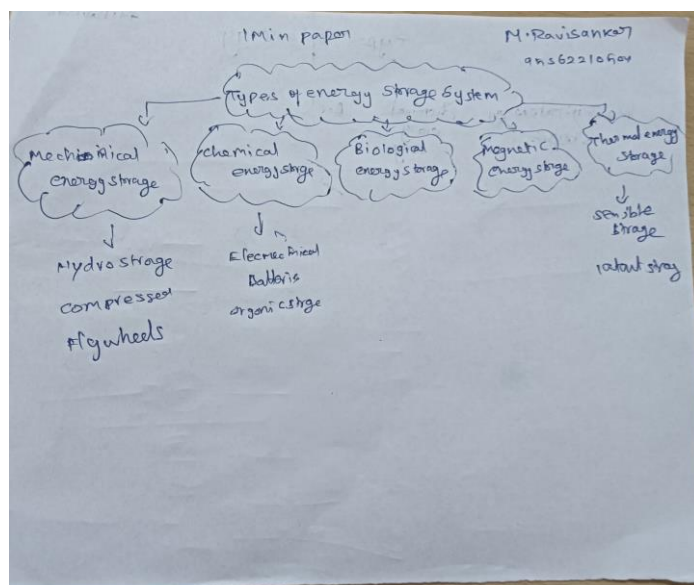
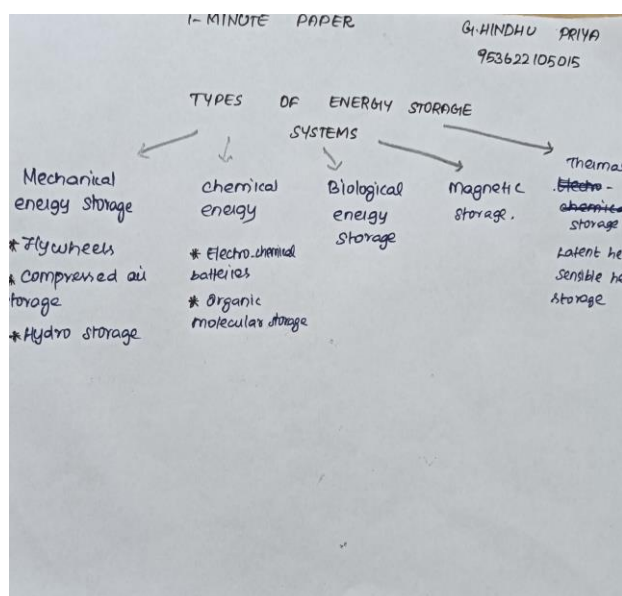
(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	PO1	PO6	PO7	PO8	PO9	PO10	PO12
	2	3	3	1	3	3	2
Justification for correlation	The basics and limitations of renewable	As role of engineering in protecting	The impact of energy Storage Systems in	Consequences of unethical environmental practice &	The students will be given group	The students will be given group assignment /	For preparing the assignment,

	energy sources are discussed for understanding the need of energy storage as an engineering problems.	the globe is discussed	environmental sustainability will be taught to the students	method to overcome the same is briefly explained	assignment / presentation	presentation. They will prepare a report on the same	students need to comprehend few technical contents
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• **Images / Screenshot of the practice:**



• **Reflective Critique:**

❖ **Feedback of practice from students and other stakeholders:**

- The assessment of effectiveness of the activity was felt when told most of the points.
- While conducting the activity, I understood that the students will be able to explain the basics of Energy Storage Systems and its various types.
- The success of the activity was evaluated by asking the same question in Internal Assessment test I – Around 90% of students answered.

Benefit of the practice:

- Students were able to attend the question even in the questions are in indirect form.
- Students were able to explain the concepts in examination without any confusion.

❖ **Challenges faced in implementation:**

- In the class mostly few students hesitate to answer the questions.
- Time utilization for conducting activity.

References:

1. Ibrahim Dincer and Mark A. Rosen, 'Thermal Energy Storage Systems and Applications', John Wiley & Sons, 3rd Edition, 2021.
2. Ru-shi Liu, Lei Zhang and Xueliang sun, 'Electrochemical technologies for energy storage and conversion', Wiley publications, 2nd Volume set, 2012.
3. James Larminie and Andrew Dicks, 'Fuel cell systems Explained', Wiley publications, 3rd Edition, 2018.

Signature of Faculty Member**HOD**



Department of Electrical and Electronics Engineering

Academic Year 2024 – 2025 (Odd Semester)

Degree, Semester & Branch: V Semester B.E. EEE

Course Code & Title: EE3032/ Energy Storage Systems

Name of the Faculty member (s): Mrs. G. Sivapriya

Innovative Practice Description

- **Unit / Topic:** Unit II / Thermal storage System , Solar Thermal Collector, Evacuated Type Flat plate collectors, Solar PV Panel
- **Course Outcome:** CO 2
- **Topic Learning Outcome:** 2a (TLO 4)
- **Activity Chosen:** Field Visit, Demonstration
- **Justification:**

Solar Photovoltaic (PV) is a technology that converts sunlight (solar radiation) into direct current electricity by using semiconductors. When the sun hits the semiconductor within the PV cell, electrons are freed and form an electric current. Solar PV technology is generally employed on a panel (hence solar panels). After teaching the constructional, I thought of conducting this as field visit for making the students to give overview of PV panel which enhance the learning level and as a teacher I can judge the understanding level of the students.

Solar thermal energy storage system in which thermal energy from solar radiation is absorbed by using solar collector and it is transferred to the fluid, finally the heated water is taken from the solar water heater. In this type of storage system, the thermal energy is directly utilised.

- **Time Allotted for the Activity:** 45 minutes

- **Details of the Implementation:**

After teaching the concept, gave students one or two minutes to think about the topic without writing anything.

Total Strength is 63,

Photographer: one student – Mr. J. M. Naveen (interested in photography)

Reporter: Myself

At the end the Class (Last 15 minutes)

- ✓ I asked the students to think about solar PV system operation and solar thermal storage system for 2 minutes.



- ✓ Then I told them to Pair with their neighbours and discuss about the operation of solar photovoltaic energy conversion system and solar thermal storage system for another 1 minute.
- ✓ Finally, I randomly asked students about the operation of solar photovoltaic energy conversion system and solar thermal storage system. (2 minutes).

• **CO – PO / PSO mapping:**

CO	PO 1	PO 2	PO 3	PO 5	PO6	PO 7	PO 8	PO 9	PO 10	PO12	PSO1	PSO2	PSO3
CO 2						2					2		
(1 – Low				2 – Moderate				3 – High)					

• **PO / PSO mapped:**

Innovative practice	PO 7
	2
Justification for correlation	The basic applications of solar energy in an environmental context are taught to the students, hence it is moderately correlated (level 2).

• **Images / Screenshot of the practice:**





Reflective Critique:

❖ *Feedback of practice from students and other stakeholders:*

Students were satisfied with the field visit demonstration class on Solar PV system and solar water heater (thermal energy storage system,) also they said, it increases our curiosity to explore more about the topic.

❖ *Benefit of the practice:*

1. Students are able to understand the impact of engineering solution on society
2. Students understood the concept which was reflected from their answers for the questions I have asked during discussion session of field visit.
3. The success of the activity was evaluated by asking the same question during IAT-I – **Around 85% of students answered.**
4. Students are able to explain the concept of thermal energy storage system in examination without any confusion.

❖ *Challenges faced in implementation:*

1. Time utilization for conducting activity became more than planned.

References:

1. D.P.Kothari, K.C Singal, RakeshRanjan “Renewable Energy Sources and Emerging Technologies”, PHI Learning Pvt.Ltd, New Delhi, 2013
2. Scott Grinnell, “Renewable Energy & Sustainable Design”, CENGAGE Learning, USA, 2016

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Department of Electrical Electronics Engineering

Academic Year 2024 – 2025(Odd Semester)

Degree, Semester & Branch: B.E/V/EEE

Course Code & Title: EE3032 Energy Storage Systems

Name of the Faculty member (s): Mrs G. Sivapriya

Innovative Practice Description

- **Unit / Topic:** Unit IV / Fuel Cell
- **Course Outcome:** CO4
- **Topic Learning Outcome:** TLO15
- **Activity Chosen:** Ungraded Quiz
- **Justification:**

I explained the working principle of fuel cell, types of fuel cell and the components of fuel cell. After teaching this topic, I thought of this activity for making to have clarity about the Fuel cell. So I can judge the level of understanding.

- **Time Allotted for the Activity:** 10 Minutes
- **Details of the Implementation:**
 - After teaching the concept, I gave students one or two minutes to think about the topic without writing anything.
 - Total Strength :63
 - Reporter: Myself
 - At the end of the Class (Last 10 minutes)
 - I asked the students to recall about the fuel cell working principle, the components of fuel cell and types of fuel cell for 2 minutes.
 - Then I told them to write to attend the quiz which is ungraded (8 minutes)
 - Finally, I collected the responses of each student in canvas.
- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO5	PO6	PO7	PO8	PO9	PO10	PO12	PSO1	PSO2
CO2	2	1	2	1	2	1	3	3	2	2	1

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	PO5	PO8
	2	1
Justification for correlation	The students are using online tool for doing this quiz. Hence it is mapped with this PO at level 2	The students are doing this quiz without discussing and copying from others. Hence it is mapped with this PO at level 1

- **Images / Screenshot of the practice:**

Student Email	Attempt	Time	Score	Grade
953622105008@ritrijm.ac.in	1	finished in 2 minutes	0	7
953622105009@ritrijm.ac.in	1	finished in 2 minutes	0	8
953622105011@ritrijm.ac.in	1	finished in 2 minutes	0	5
953622105015@ritrijm.ac.in	1	finished in 4 minutes	0	6
953622105021@ritrijm.ac.in	1	finished in 4 minutes	0	8
953622105022@ritrijm.ac.in	1	finished in 2 minutes	0	8
953622105027@ritrijm.ac.in	1	finished in 2 minutes	0	8
953622105029@ritrijm.ac.in	1	finished in 4 minutes	0	8
953622105031@ritrijm.ac.in	1	finished in 5 minutes	0	6
953622105033@ritrijm.ac.in	1	finished in 8 minutes	0	8
953622105035@ritrijm.ac.in	1	finished in 5 minutes	0	8
953622105036@ritrijm.ac.in	1	finished in 2 minutes	0	8
953622105048@ritrijm.ac.in	1	finished in 2 minutes	0	9

Reflective Critique:

❖ *Feedback of practice from students and other stakeholders:*

I preferred this activity because it will help me to evaluate what types of questions I am asking students and what level they are working at.

Benefit of the practice: (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)

The students can understand their knowledge in that particular topic.

❖ *Challenges faced in implementation:*

- Since it is conducted online ,it is very tough to make all students to attend.
- It is difficult to access whether student have attended the quiz properly or not.

References:

- ❖ https://www.nactateachers.org/images/The_Student-Developed_Quiz_or_Exam-Scaffolding_Higher-Order_Thinking.pdf.
- ❖ James Larminie and Andrew Dicks, 'Fuel cell systems Explained', Wiley publications, 3rd Edition, 2018.
- ❖ <https://www.netl.doe.gov/sites/default/files/netl-file/FCHandbook7.pdf> Fuel Cell Handbook

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Department of Electrical and Electronics Engineering
Academic Year 2024 – 2025 (Odd Semester)

Degree, Semester & Branch: V Semester B.E. EEE

Course Code & Title: EE3032 Energy Storage Systems

Name of the Faculty member: Mrs. G. Sivapriya, AP/EEE

Innovative Practice Description

- **Unit / Topic:** Unit V / Types of alternate Energy Storage Technologies
- **Course Outcome:** CO 5
- **Topic Learning Outcome:** TLO18
- **Activity Chosen:** Mind - Map
- **Justification:**
 - Explain the different types of alternate Energy Storage Technologies
 - After teaching the concept, I thought of conducting this activity for making the students to give the difference between the 3 types of Energy Storage Technologies which enhance the learning level and as a teacher I can judge the understanding level of the students.

Time Allotted for the Activity: 6 Minutes

After teaching the concept, the students were made to pair with their neighbors

Photographer: Myself

Reporter: Myself

At the end the Class (Last 6 minutes)

- ✓ I asked the students to think about types of Energy Storage Technologies concept for 2 minute.
- ✓ Then I told them to Pair with their neighbors and discuss about the concepts for another 1 minute.
- ✓ Finally I told them to design mind-map and submit within 3 minutes.

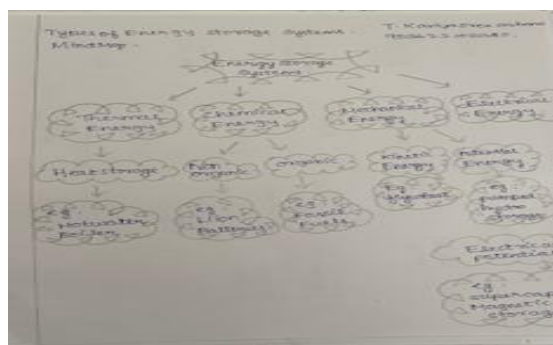
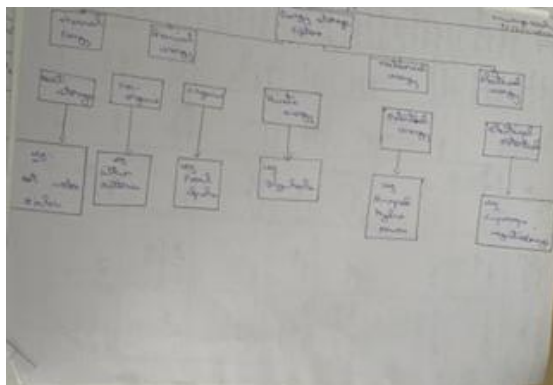
- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO5	PO6	PO7	PO8	PO9	PO10	PO12	PSO1	PSO2	PSO3
C215.2	3	2	3	2	1	2	1	2	2	2	2	1	1

- PO / PSO mapped:

Innovative practice	PO9	PO10
	2	2
Justification for correlation	The students are working in a group to list out the points and design a mind map. Hence it is mapped with level 2.	The students are drawing a neat mind map and trying to incorporate as much as points possible. Hence their visualization skills and presentation skills are improved. It is mapped with level 2.

- Images / Screenshot of the practice:



Reflective Critique:

1. Pre-implementation Reflection:

I preferred this Activity because they types of Energy Storage Technologies separately.

Challenges anticipated:

- In the class few boys hesitate to answer to the questions.
- Time utilization for conducting activity.

Steps taken:

- The boys are sitting in 2 columns – I planned to choose more pairs from boys to involve them in the activity.

2. Post implementation Reflection:

Benefits:

- Students understood the concept which was reflected from their answers for the questions I have asked during discussion session.

Challenges:

- Slow learners were not able to understand some topics during discussion
- ❖ *Benefit of the practice:* (E.g.: Outcome attainment would have increased due to innovative practice over conventional practice)
- ✓ The assessment of effectiveness of the activity was felt when told most of the points.
 - ✓ While conducting the activity, I understood that the students will be able to explain the concepts of types of Energy Storage Technologies.
 - ✓ The success of the activity was evaluated by asking the same question in Internal Assessment test II – Around 85% of students answered.

References:

1. <https://omerad.msu.edu/teaching/teaching-strategies/active-learning-strategies/27-teaching/184-visual-modeling-mini-maps>

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